

CS 348 Milestone 1

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PokePals Description

This project aims to create a database-driven web application for Pokemon card trading targeted to Pokemon enthusiasts. We plan to use the dataset "[Pokemon TCG All Cards 1999 - 2023](#)" from Kaggle, which includes information about all of the Pokemon cards released over the years. This dataset provides attributes including the card names, rarity, release date, generation, and more.

On the application, users will be able to create/login to their own profile and search or filter through available cards by their attributes. If the user sees a Pokemon card that they really like, they can request to trade cards with the owner of that Pokemon card. Users will also be able to create their own cards to add to their collection as well. We, as the developers, are the administrators of the database and we will also select some diehard Pokemon fans as well.

Planned Features

- Gift and request Pokemon cards to/from other users
- Trade Pokemon cards with other users
- Upload and create new and personalized Pokemon cards
- Filter and search for Pokemon cards in the home page
- The ability to log in, sign up, and allocate dedicated profile for each user

Trade a Pokemon card

The user can trade their Pokemon cards with other users. They would receive a Pokemon card and the other user will also receive a Pokemon card.

Query template:

```
INSERT INTO transaction (transaction_id, card_id, sender_id, receiver_id, date) VALUES  
([card_id value], [sender_id value], [receiver_id value], [date value]);  
INSERT INTO Transaction (transaction_id, card_id, sender_id, receiver_id, date) VALUES (  
[card_id value], [reciever_id value], [sender_id value], [date value]);
```

Sample query:

```
INSERT INTO transaction (card_id, sender_id, receiver_id, date) VALUES ('bare-283', 'user-32',  
'user-33', NOW());  
INSERT INTO Transaction (card_id, sender_id, receiver_id, date) VALUES ('bare-555',  
'user-33', 'user-32', NOW());
```

Expected Output:

transaction_id	card_id	sender_id	reciever_id	date
22	'bare-283'	'user-32'	'user-33'	2025-02-07 12:34:56
23	'bare-555'	'user-33'	'user-32'	2025-02-07 12:34:56

Create a new user

To begin using the site, users can create an account. They can login to the account to access their cards.

Query template:

```
INSERT INTO Account (uid, username, password, bio, pfp) VALUES ([uid], [username], [password], [bio], [pfp]);
```

Sample query:

```
INSERT INTO account (uid, username, password, bio, pfp)
VALUES (123456789, 'cool_username', 'password', 'New user', 'https://coolphoto.com/pfp');
```

Expected Output:

uid	username	password	bio	pfp
123456789	'cool_username'	'password'	'New user'	'https://coolphoto.com/pfp'

Create new Pokemon card

The users can make their own Pokemon cards which can then be traded.

Query template:

```
INSERT INTO `Pokemon Card` (card_id, pname, set_name, is_custom, image_url, generation, release_date, rarity, subtype, hp, level, flavour_text)
VALUES ([card_id], [pname], [set_name], TRUE, [image_url], [generation], [release_date], [rarity], [subtype], [hp], [lvl], [flavour_text]);
```

Sample query: *INSERT INTO pokemon_card (card_id, pname, set_name, is_custom, image_url, generation, release_date, rarity, subtype, hp, level, flavour_text)*
VALUES ('Poke123', 'Pikachu', 'Base Set', TRUE, 'https://example.com/pikachu.jpg', 'Generation I', 1999-01-09, 'Rare', 'Electric', 60, 12, 'A mouse Pokémon');

Expected output:

card_id	pname	set_name	is_custom	image_url	generation	release_date	rarity	subtype	hp	level	flavour_text
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'Poke 123'	'Pikachu'	'Base Set'	TRUE	'https://example.com/pikachu.jpg'	'Generation I'	1999-01-09	'Rare'	'Electric'	60	12	'A mouse Pokémon'
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Search and filter for Pokemon cards

Users can search for Pokemons based on their attributes.

Query template: *SELECT * FROM pokemon_card WHERE [add queried attributes here];*

Sample template: *SELECT * FROM pokemon_card WHERE pname = 'Pikachu';*

Expected output:

card_id	pname	set_name	is_custom	image_url	generation	release_date	rarity	subtype	hp	level	flavour_text
'Poke 123'	'Pikachu'	'Base Set'	TRUE	'https://example.com/pikachu.jpg'	'Generation I'	1999-01-09	'Rare'	'Electric'	60	12	'A mouse Pokémon'

Description of Platform

We will be hosting locally and users will interact through a TypeScript React web application front end and Python backend which will connect to a MySQL database.

Sample Plan

Data will be populated through *mysqlimport*, a data import platform by MySQL which provides a command-line interface to the LOAD DATA SQL statement. Through this, we can import the CSV file of our data into MySQL.

Assumptions

- 1) There is only one copy of every Pokemon card in existence

- 2) All cards are Pokemon. There are no energy, trainer, etc. cards.
- 3) Pokemon can belong to only one or two types
- 4) The pokemon id from Kaggle dataset (pokemon-tcg-data-master 1999-2023) is unique
- 5) The pokemon id from Kaggle dataset (pokemon-tcg-data-master 1999-2023) has a one-to-one match for the id from the Kaggle dataset (Pokemon Cards)
Ex: a Pokemon with id: X from **pokemon-tcg-data-master 1999-2023** would be the exact same card as Pokemon with id: X from **pokemon cards**
- 6) Users are trading virtual Pokemon cards
- 7) Every Pokemon card only has one ability. But two or more Pokemon cards can have the same ability.
- 8) Pokemon Weakness and Resistances follows this chart:
- <https://www.thegamer.com/pokemon-types-weakness-chart/>
- 9) The damage attribute in attack is the minimum damage a Pokemon can do. We will not consider actual damage because that is dependent on gameplay and non-Pokemon cards which is not in our assumption that all cards are Pokemon cards.

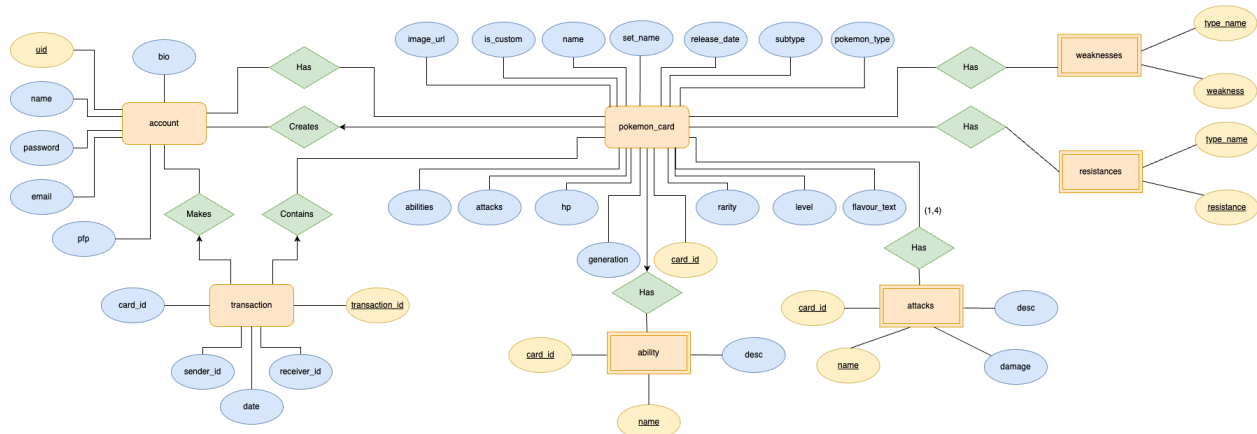
E/R Diagram

Please zoom in to view

Link to diagram

Also available in github repo under milestone 1

https://drive.google.com/file/d/1Fe7H5xObP88uInN23j_kFt-Wdr700rpF/view?usp=sharing



List of Tables

Account

Attribute	Type	Constraints
uid	DECIMAL (9,0)	Primary Key, auto increment

username	VARCHAR	
email	VARCHAR	
usr_password	VARCHAR	
bio	VARCHAR	
pfp	VARCHAR	

Transaction

Attribute	Type	Constraints
tid	DECIMAL (9,0)	Primary Key, auto increment
card_id	VARCHAR	Foreign key to Pokemon Card(card_id)
sender_id	DECIMAL (9,0)	Foreign key to Account(uid)
receiver_id	DECIMAL (9,0)	Foreign key to Account(uid)
tdate	DATETIME	

Pokemon Card

Attribute	Type	Constraints
card_id	VARCHAR	Primary Key
pname	VARCHAR	
set_name	VARCHAR	
is_custom	BOOLEAN	
image_url	VARCHAR	
generation	VARCHAR	
release_date	DATE	
rarity	VARCHAR	
pokemon_type	VARCHAR	

subtype	VARCHAR	
hp	INT	
level	INT	
flavour_text	VARCHAR	

Abilities

Attribute	Type	Constraints
card_id	VARCHAR	Foreign key to Pokemon Card(card_id)
ability_name	VARCHAR	
description	VARCHAR	

Primary key: (card_id, ability_name)

Attacks

Attribute	Type	Constraints
card_id	VARCHAR	Foreign key to Pokemon Card(card_id)
attack_name	VARCHAR	
description	VARCHAR	
damage	INT	

Primary key: (card_id, attack_name)

Weaknesses

Attribute	Type	Constraints
type_name	VARCHAR	Foreign key to Pokemon Card(pokemon_type)
weakness	VARCHAR	

Primary key: (type_name, weakness)

Resistances

Attribute	Type	Constraints
type_name	VARCHAR	Foreign key to Pokemon Card(pokemon_type)
resistance	VARCHAR	

Primary key: (type_name, resistance)

Members Contributions

Contributions of members for Milestone 1:

- Rosanne - contributed to report document and ideas for ER diagram and schema; designed the frontend of the application
- Sabrina - contributed to report document and ideas for ER diagram and schema
- Siting - contributed to report document and ideas for ER diagram and schema
- Jessica - contributed to report document and ideas for ER diagram and schema
- Crystal - contributed to report document and ideas for ER diagram and schema

GitHub Repo

<https://github.com/sabrina-xing/pokemon>